Harrison Pielke-Lombardo

Senior Backend & Full-Stack Engineer | Available for Contract & 1099 Work

Denver, CO (remote or hybrid) | 720-209-6249 | pielkekid@gmail.com | linkedin.com/in/tuh8888 | github.com/tuh8888

SUMMARY

Senior backend and full-stack engineer available for contract and 1099 engagements, remote or on-site in Colorado. 10 years of building production systems in Python, Clojure, TypeScript, and Java. Most recently built the document-processing pipeline, rules engine, and AWS infrastructure behind a B2B SaaS logistics platform handling 1M+ documents per month, which grew from zero to $2.1M ARR and was acquired. Comfortable owning a project end to end, working directly with clients, and getting productive quickly in an existing codebase.

WHAT I CAN DELIVER

Backend APIs and data pipelines: REST and GraphQL services, high-volume document and data processing, rules engines

Database migrations: zero-downtime production migrations, including Datomic to PostgreSQL at Datomic's size limits

AWS infrastructure: Terraform-managed, event-driven systems on Lambda, ECS, SQS, SNS, and RDS

LLM and ML automation: production document processing and verification that replaces manual review work

Clojure and ClojureScript: new builds, maintenance, and modernization of existing Clojure systems

SKILLS

Languages: Python, Clojure, ClojureScript, TypeScript, Java, JavaScript, SQL, Bash

Frontend: TypeScript, ClojureScript, React (Reagent / re-frame), JavaScript, data visualization

Data: PostgreSQL, Datomic, DynamoDB, Redis, SQL, knowledge graphs (RDF)

Cloud & DevOps: AWS (Lambda, ECS, EC2, SQS, SNS, RDS, DynamoDB, CloudFront), Terraform, Docker, CI/CD, Git, GitHub

AI & ML: LLM integration, PyTorch, NLP, AI-assisted development (Claude, Copilot)

EXPERIENCE

Senior Software Engineer | Sep 2022 - Mar 2026

Peruse Technology Inc. (acquired), Remote

- Helped scale a B2B SaaS logistics platform from zero to $2.1M ARR and 70+ business customers as part of a 4-6 person engineering team, leading to the company's acquisition.
- Designed and built the core load processing pipeline and a custom rules engine in Clojure and Python, processing 1M+ documents per month through REST and GraphQL APIs.
- Trained and deployed LLM-based models in PyTorch to automate document processing and freight broker verification, replacing work that took customers entire back-office audit teams.
- Planned and executed the zero-downtime migration of the production database from Datomic to PostgreSQL as it approached Datomic's size limits.
- Shared ownership of event-driven AWS infrastructure as code in Terraform (Lambda, ECS, SQS, SNS, RDS); took on-call rotations and contributed to the TypeScript frontend.
- Served as technical contact on client calls and owned relationships with several client accounts.

Research Software Engineer (Graduate Researcher) | Aug 2016 - Aug 2022

University of Colorado Anschutz Medical Campus, Aurora, CO

- Led software for interconnected research projects across 4+ cross-functional teams, shipping 5 open-source package releases, 4 dataset releases, and 6 peer-reviewed publications.
- Built Knowtator, a Java and ClojureScript / re-frame web app for text annotation and ontology development (rewritten from a Java Swing desktop app), used to create 100k+ annotations by clinicians and researchers.
- Improved relation-extraction precision by 10% with a novel bootstrapping algorithm accelerated with GPU matrix operations.

OPEN SOURCE & SPEAKING

- datascript-shortest-path, a graph library with a Fibonacci heap built on immutable data structures; contributor to the re-frame ecosystem (day8/re-frame-undo).
- Invited speaker, "AI in the Transportation Sector: Challenges and Successes," Institute for Data, Econometrics, Algorithms, and Learning (IDEAL), Chicago, June 2024.

EDUCATION

M.S., Computational Bioscience, University of Colorado Anschutz Medical Campus | 2022

B.S., Applied Mathematics (minors: Computer Science, Biochemistry), University of Colorado Boulder | 2016